KPR Institute of Engineering and Technology
Learn Beyond
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PROJECT PROGRESS REPORT

Faculty Management System

Features developed so far, current status and remaining work

PREPARED FOR	KPR Institute of Engineering and Technology, Coimbatore
REPORT DATE	26 October 2026
PROJECT STATUS	Core development complete; ready for deployment set-up
ENVIRONMENT	Running in development (not yet deployed)

1. Summary

The Faculty Management System (FMS) brings faculty records, research, attendance, leave, appraisals, approvals and accreditation reporting for KPRIET into one web portal. Faculty enter information once, their HOD verifies it online, and the same verified records feed dashboards, appraisals and NAAC/NBA/NIRF reports.

All core modules are developed and working end to end in the development environment, with role-based access for five roles, a refreshed user interface, an AI assistant and strong account security. The main remaining steps are deployment on a college server, connecting the college email system, loading real data and user acceptance testing.

5 User roles	**14** Functional modules	**61** Screens
410 API endpoints	**36** Automated test suites	**6** AI features

All 36 automated test suites pass on the latest version.

Status at a glance

Area	Status	Notes
Faculty profile, research, activities, students	● Completed	Data entry, verification by HOD, history
Attendance, leave and appraisals	● Completed	Multi-level approvals HOD → Dean → Principal
Dashboards, analytics and reports	● Completed	Role dashboards with live charts; accreditation reports
Administration and bulk import	● Completed	Users, departments, calendar, settings, Excel import, audit log
Account security	● Completed	Two-step sign-in, signed-in devices, password reset, audit trail
AI assistant (Google Gemini)	● Completed	Six features; uses a development key for now
User interface redesign	● Completed	New theme, home and sign-in pages, animations, mobile layout
Email delivery	● Needs setup	Built and tested; needs the college mail server details
Teaching module (subjects, workload, timetable)	● Built, hidden	Hidden at the client's request; can be switched on
Production deployment	● Pending	Needs a server, web address and HTTPS
Real data and user acceptance testing	● Pending	After deployment

2. Features developed

Every module follows the same pattern: faculty add a record with proof, the HOD verifies or returns it with a comment, and verified records appear in the profile, the appraisal and the reports. Every action is written to the audit log.

2.1 Roles and access

- Five roles: **Faculty**, **Head of Department**, **Dean**, **Principal** and **Administrator**, each with its own dashboard and menu.
- Access is enforced on the server: HODs see only their department; faculty see only their own records.
- Leadership can preview other roles' dashboards.

2.2 Faculty profile and directory

- Complete profile: personal details, qualifications, experience, awards and memberships.
- Private document vault for certificates and evidence, with preview and secure download.
- Profile verification by the HOD; faculty directory with search and filters.

2.3 Research and intellectual property

- Publications (journals, conferences, books, chapters) with indexing, quartile, citations and PDF upload.
- **Auto-fill from DOI** (Crossref) and **import from ORCID**, so faculty do not retype paper details.
- Funded projects with sanctioned and received amounts and documents; patents, copyrights and consultancy.

2.4 Academic activities

- Conferences, workshops, seminars, FDPs and STTPs with certificates.
- **AI certificate reading:** upload a certificate and the form fills in the title, organiser, dates and duration.

2.5 Students

- Student mentoring with meeting records, action plans and outcomes; student projects; PhD scholars with stage tracking.
- **AI mentoring insights** flag students who may need help, with reasons and suggested next steps.

2.6 Attendance and leave

- Daily and monthly attendance, correction requests, and locking of finished months.
- Leave applications with balances, calendar view and multi-level approval; cancellation with reason.

2.7 Performance appraisal

- Appraisal cycles with categories and weightage; self-appraisal with evidence.
- Review by HOD, then Dean, then Principal, with scores, comments and final grade.
- **AI drafting** of the teacher's remarks from verified records, and an **AI summary** for reviewers.

2.8 Approvals

- One approval queue for publications, projects, activities, IPR, leave and attendance corrections.
- Approve, return for correction with comments, or reject; full approval history.
- **AI check against proof:** compares what was entered with the uploaded certificate or paper and flags mismatches and duplicates.

2.9 Dashboards, analytics and reports

- Role dashboards with live charts built from real data: publications by year and quartile, attendance trends, pending items, department comparison, research output and grants, patent progress.
- Institution analytics, faculty comparison, and HOD, Dean and Principal consoles.
- Accreditation reports (NAAC, NBA, NIRF) and data export; **AI draft of NAAC Criterion III** from live data.
- **Ask the data:** leadership can ask questions in plain English, for example "How many Q1 papers did ECE publish this year?"

2.10 Administration

- User and role management, departments, programmes, academic years and semesters, master data and institution settings.
- **Bulk import of faculty from Excel** with a downloadable template and a row-by-row error preview; each new account receives a set-password email.
- Self-registration with college email, confirmed by an emailed code and approved by an administrator.
- Complete audit log of sign-ins, changes and approvals.

2.11 Notifications and calendar

- In-app notifications that appear instantly (live updates), with email for approvals and leave decisions; each user can choose what to receive.
- Private calendar link so leave and timetable appear in Google Calendar, Outlook or a phone calendar.

2.12 Account security

- Account lockout after repeated wrong passwords; password reset by email link; password rules.
- **Two-step sign-in** with an authenticator app and recovery codes (recommended for leadership).
- **Signed-in devices** list with remote sign-out; changing a password signs out other devices.
- Protection against common web attacks (CSRF, injection, unsafe file uploads), request limits, and private storage of documents.

2.13 User interface

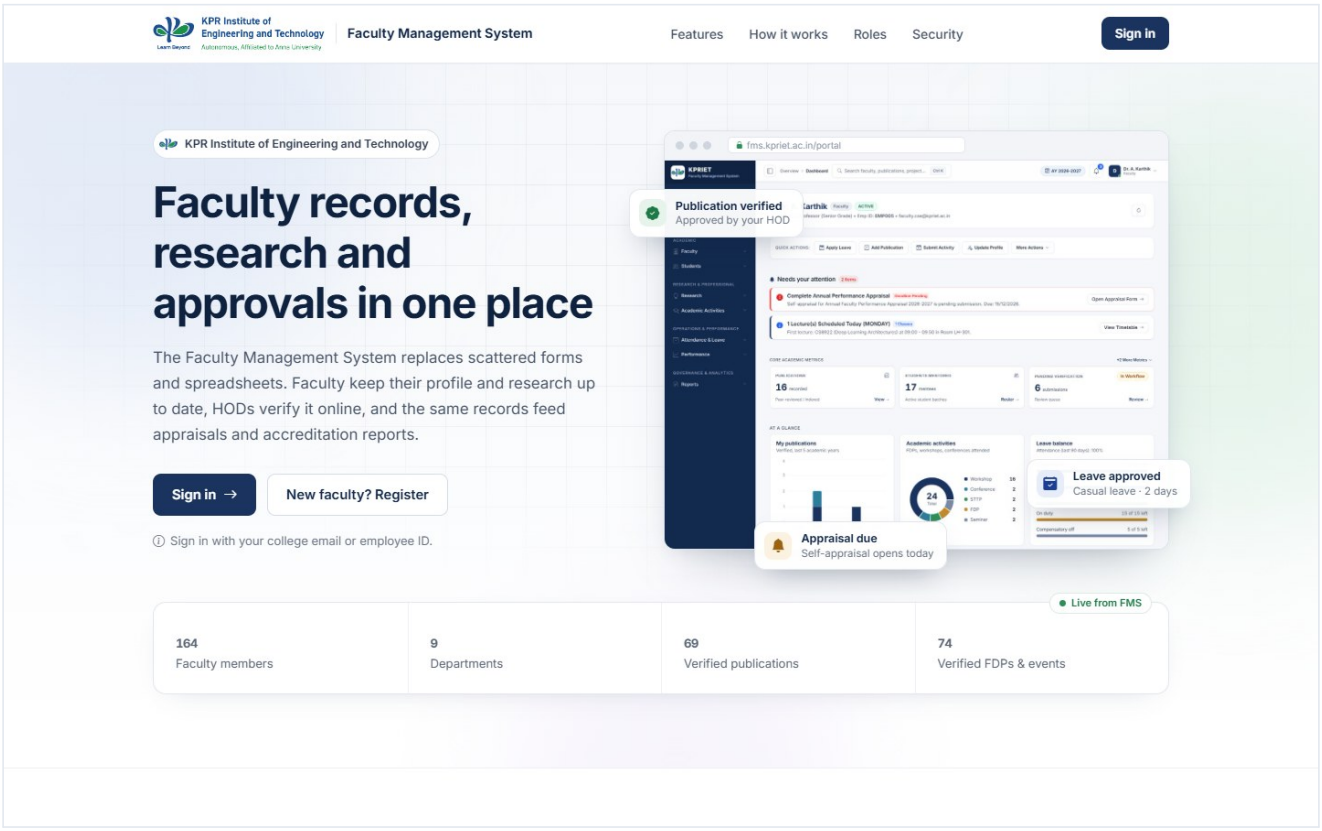
- Redesigned, professional theme across the portal; new home page and sign-in pages.
- Animated charts, counters and hover effects, with a calmer version for users who turn animations off.
- Works on phones and tablets; can be installed on a phone like an app.

2.14 Teaching module (built, currently hidden)

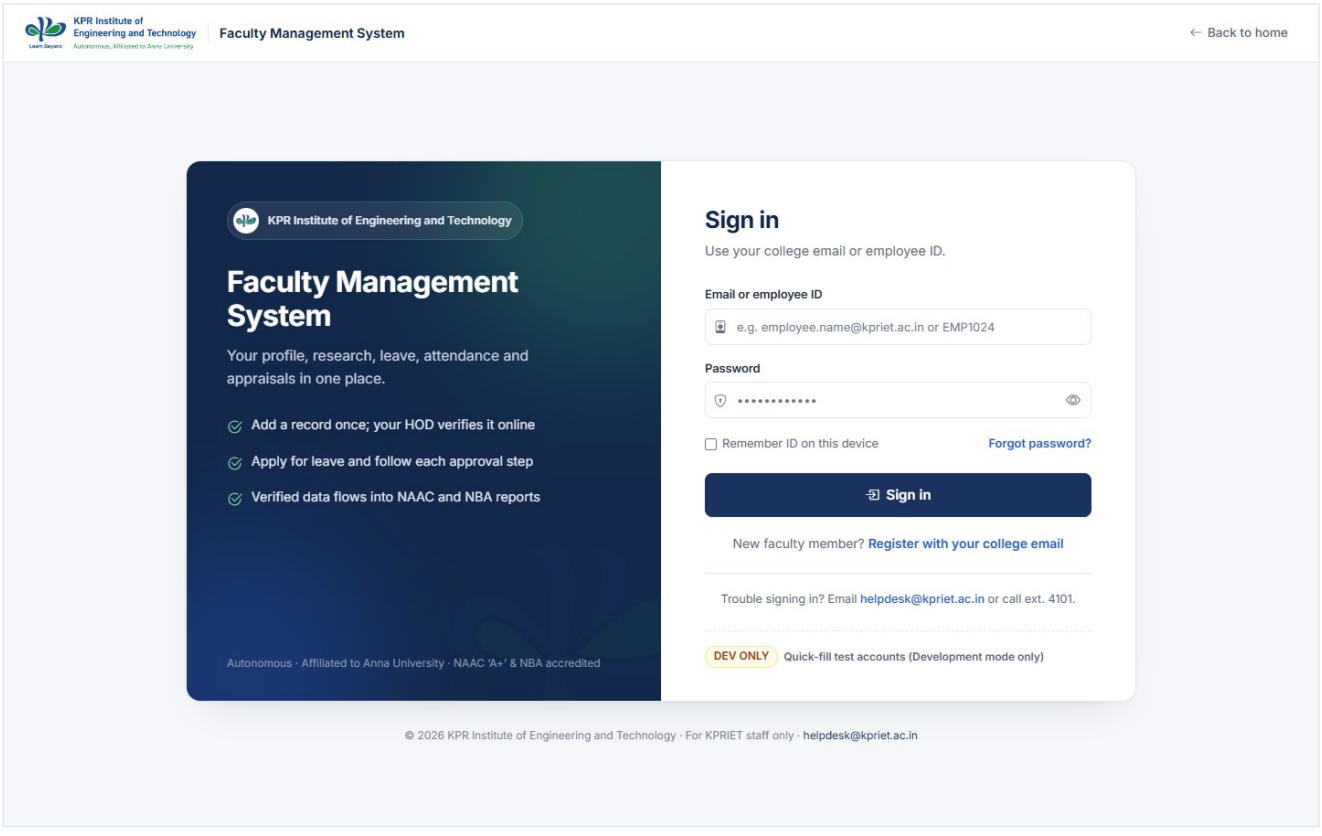
- Subjects, faculty allocation, workload with thresholds, timetable with clash detection, and substitutions.
- Hidden from all menus at the client's request; it can be switched back on without further development.

3. Selected screens

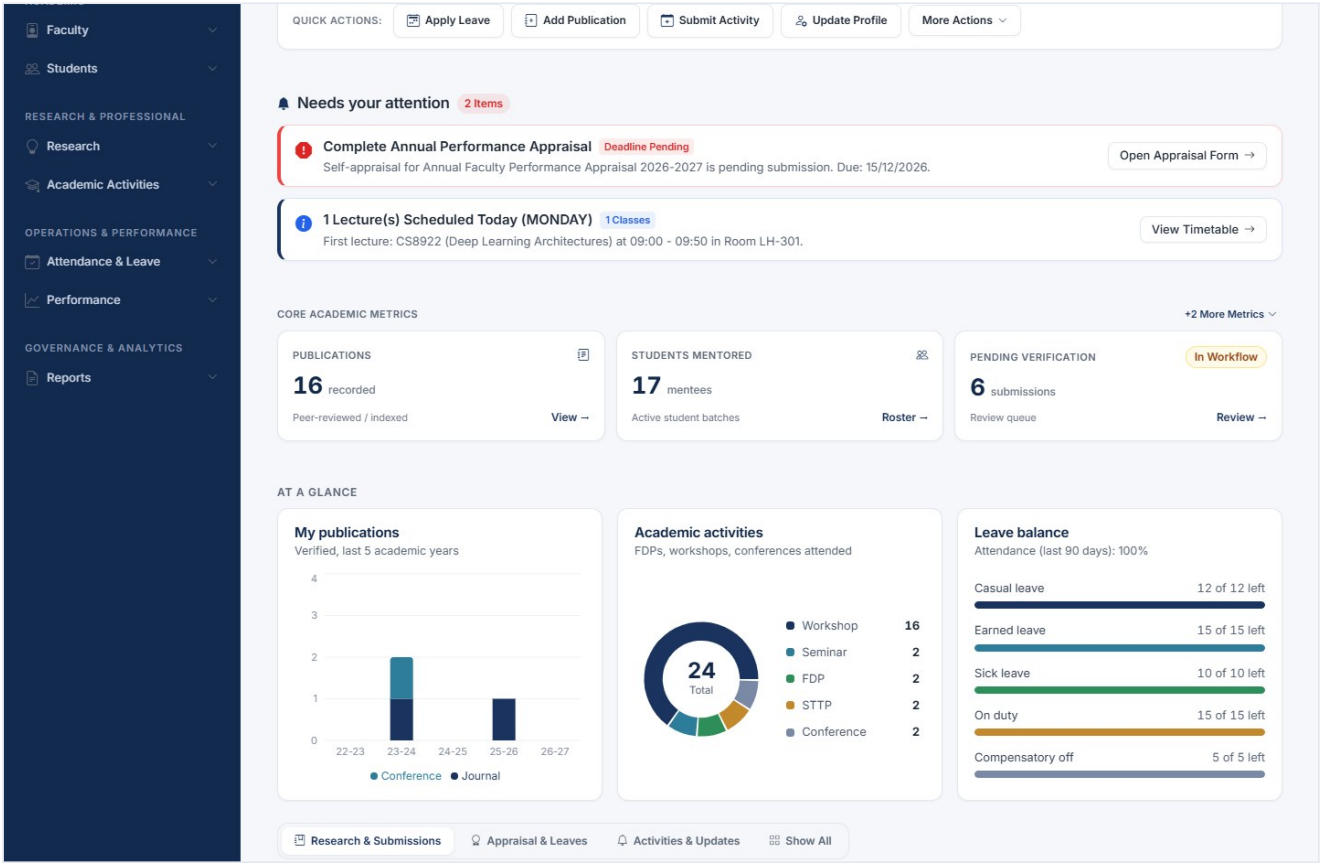
Screenshots from the development environment using sample data (names and figures are test data).



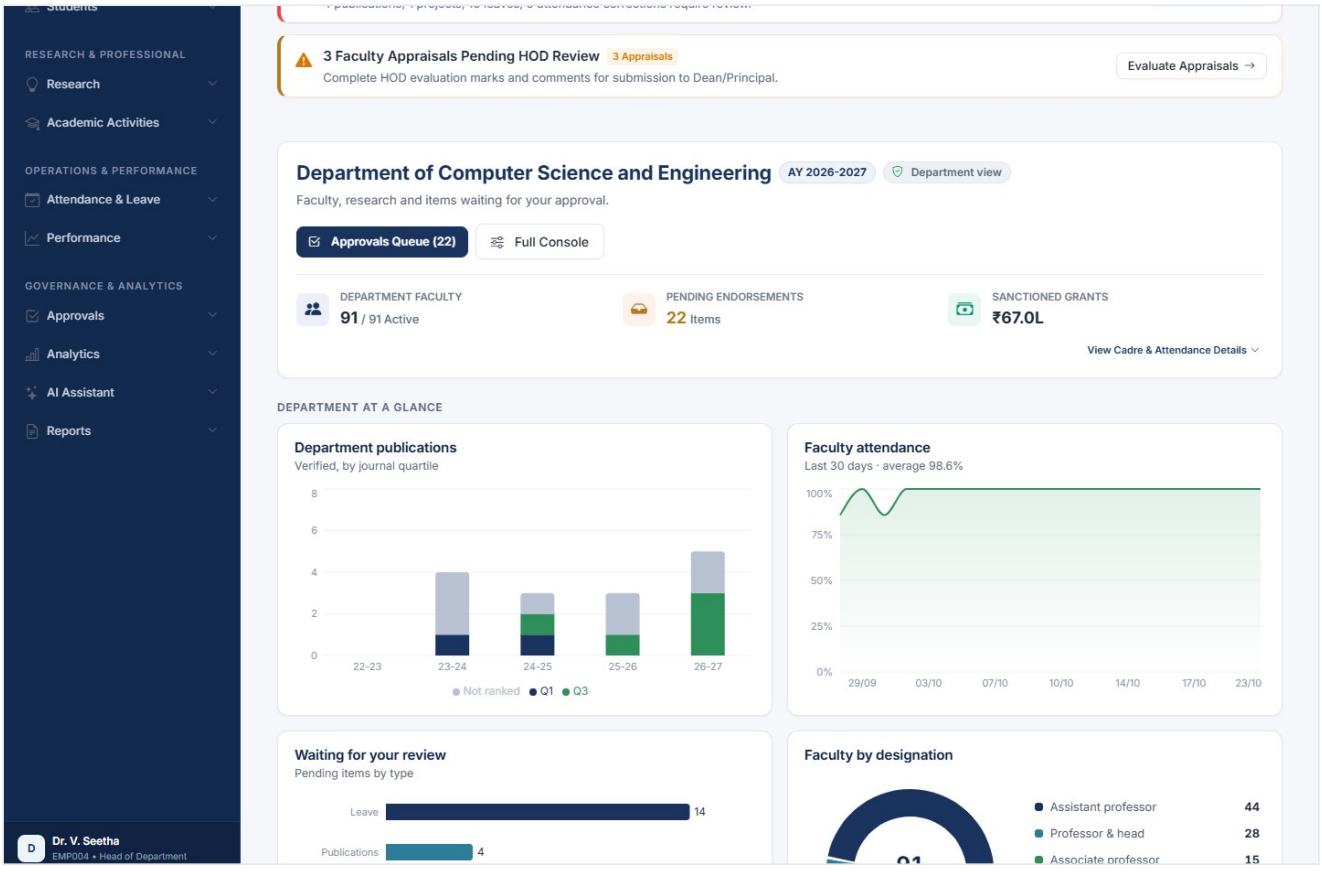
Public home page with live figures from the system



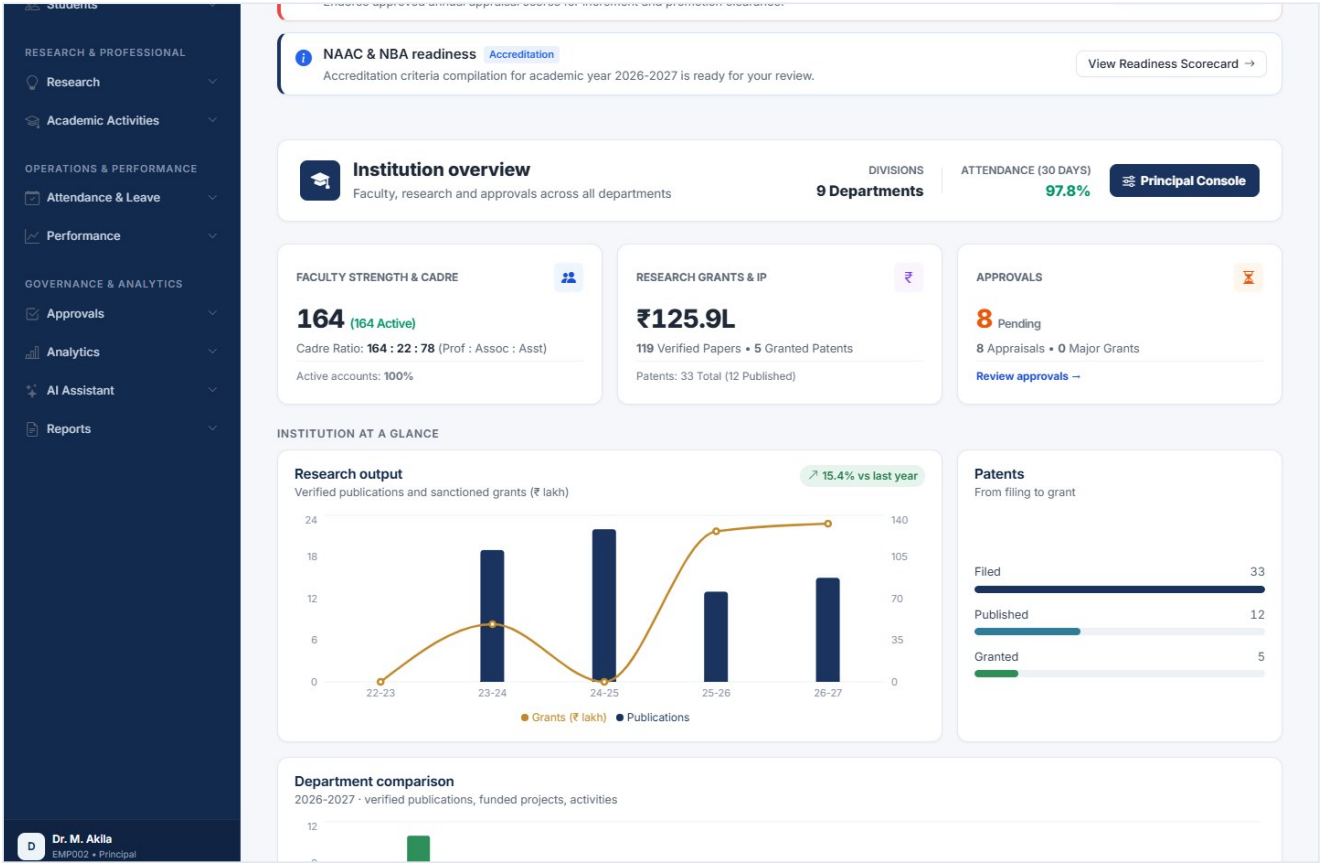
Sign-in page (two-step sign-in and password reset supported)



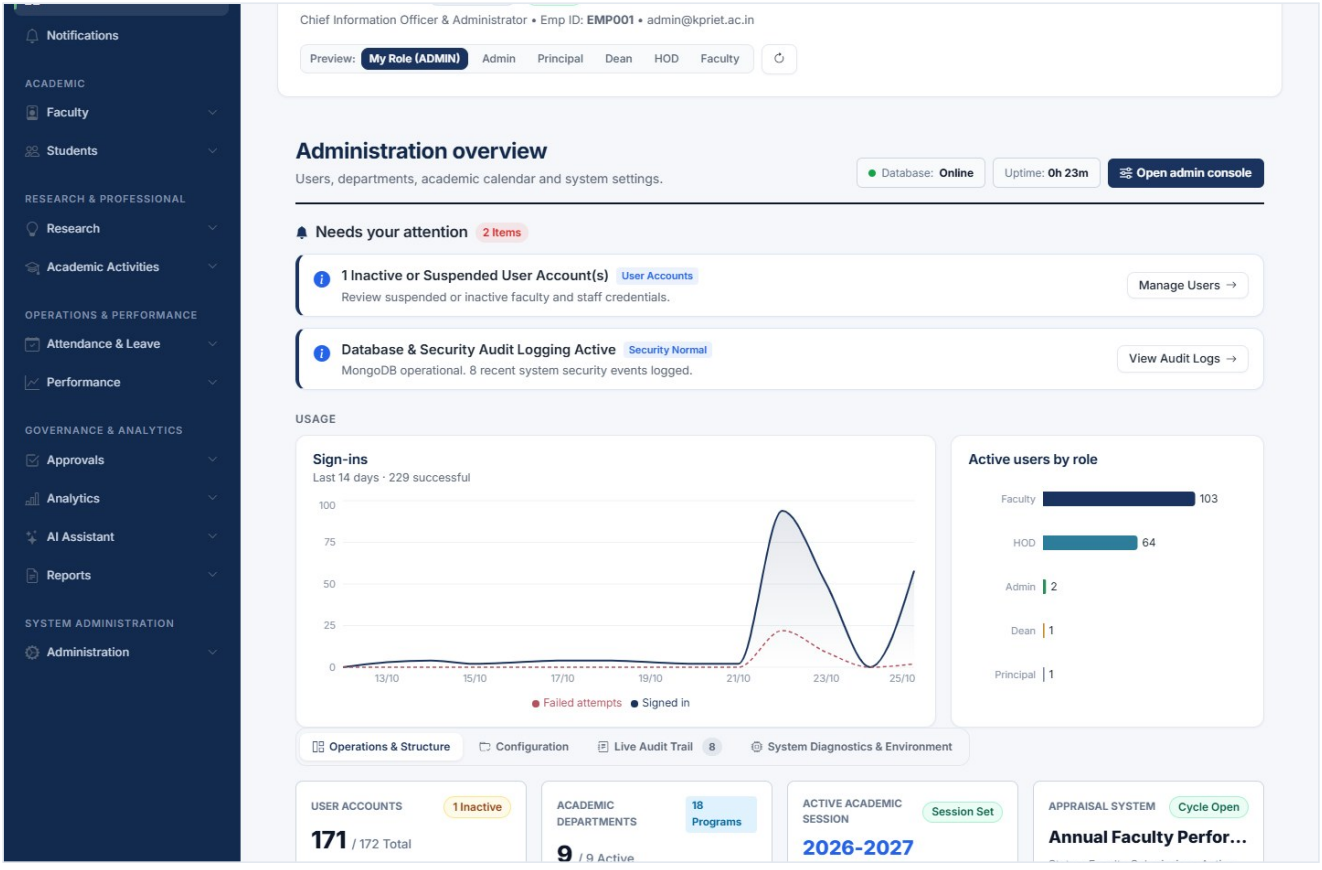
Faculty dashboard: attention items, key figures and personal charts



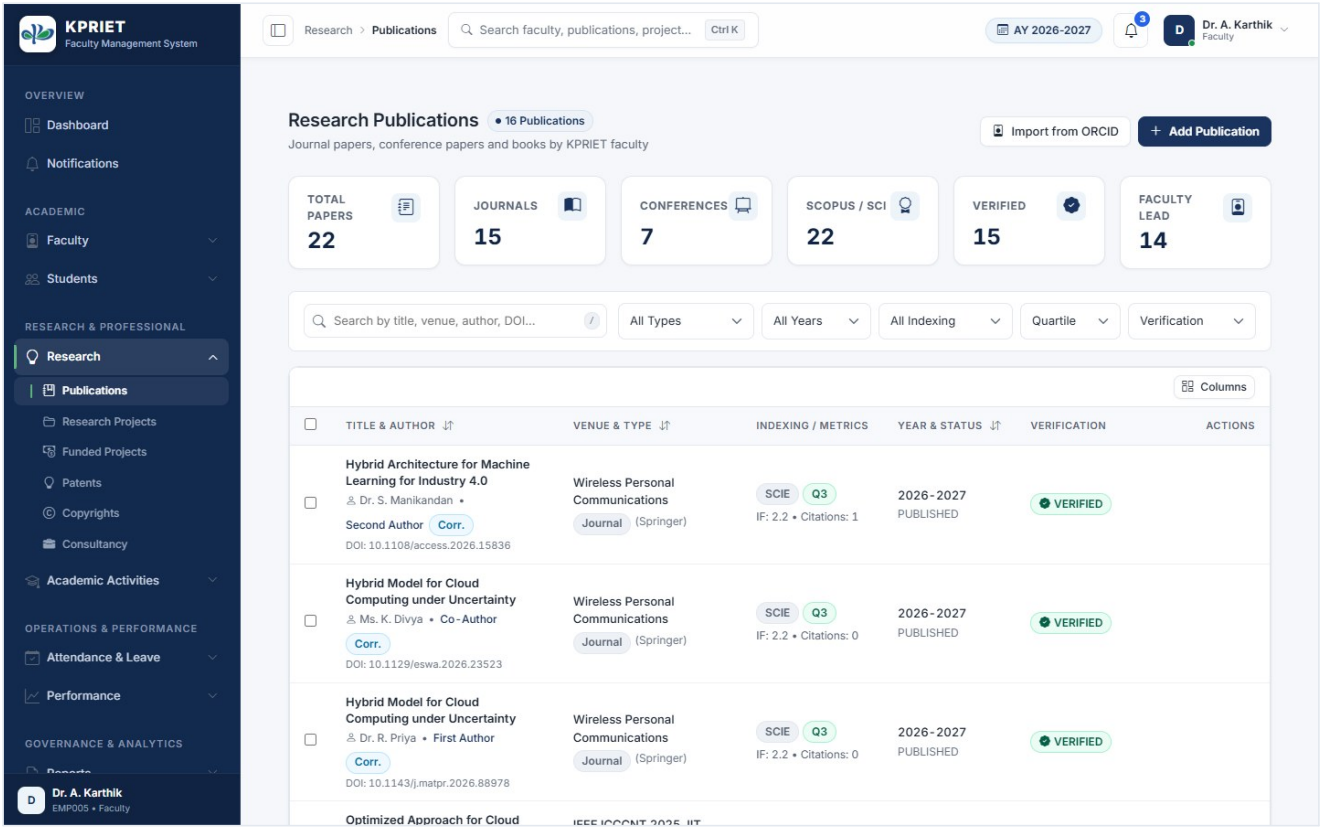
HOD dashboard: department publications, attendance trend and pending reviews



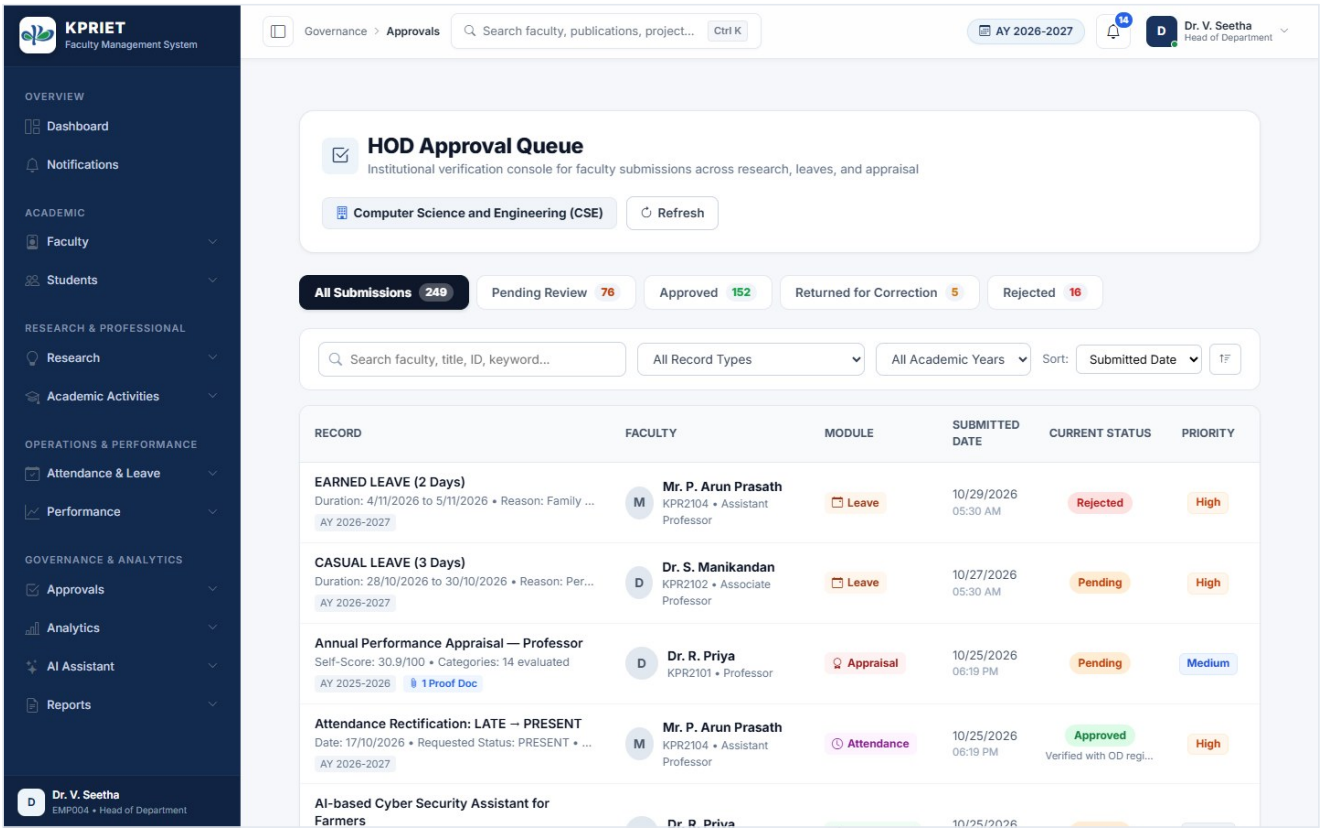
Principal dashboard: research output, department comparison and patents



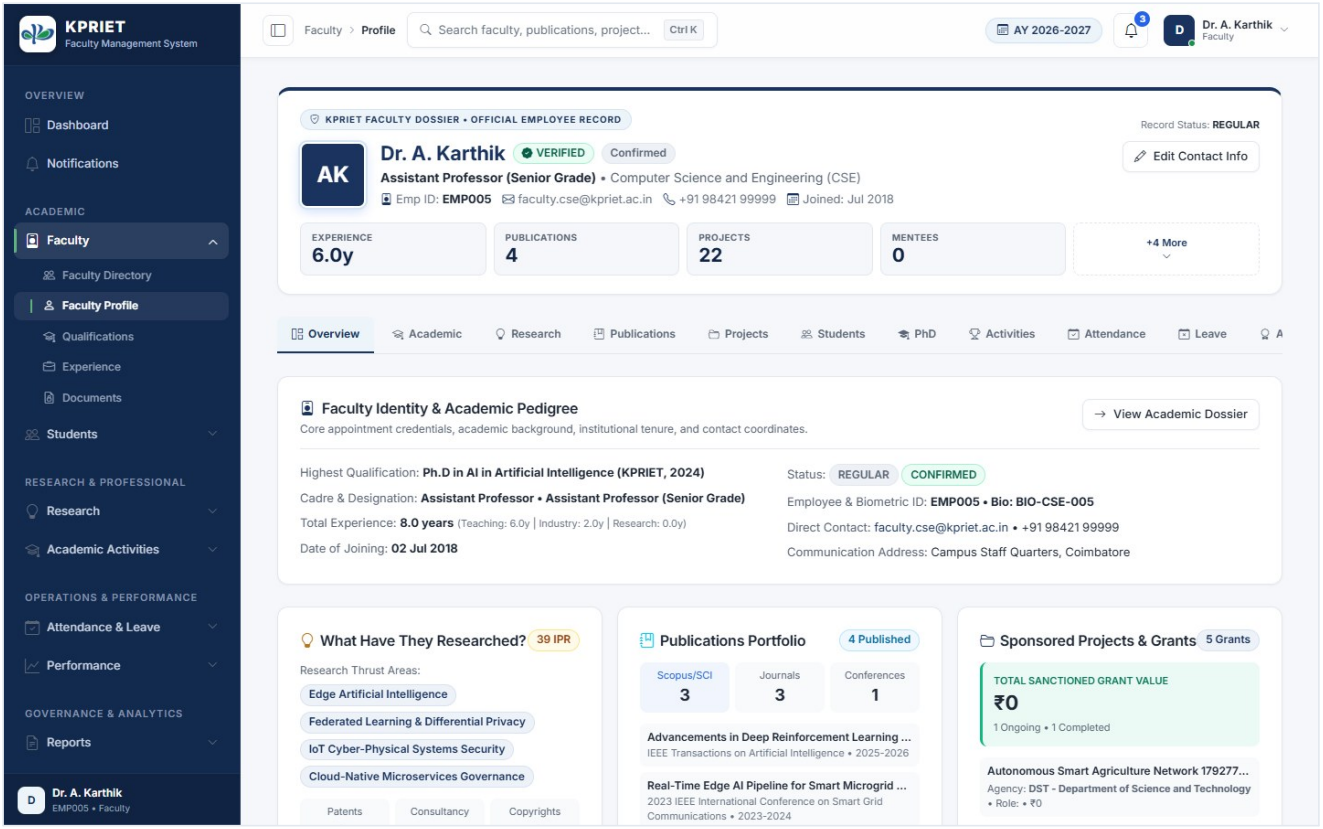
Administrator dashboard: usage charts and system configuration



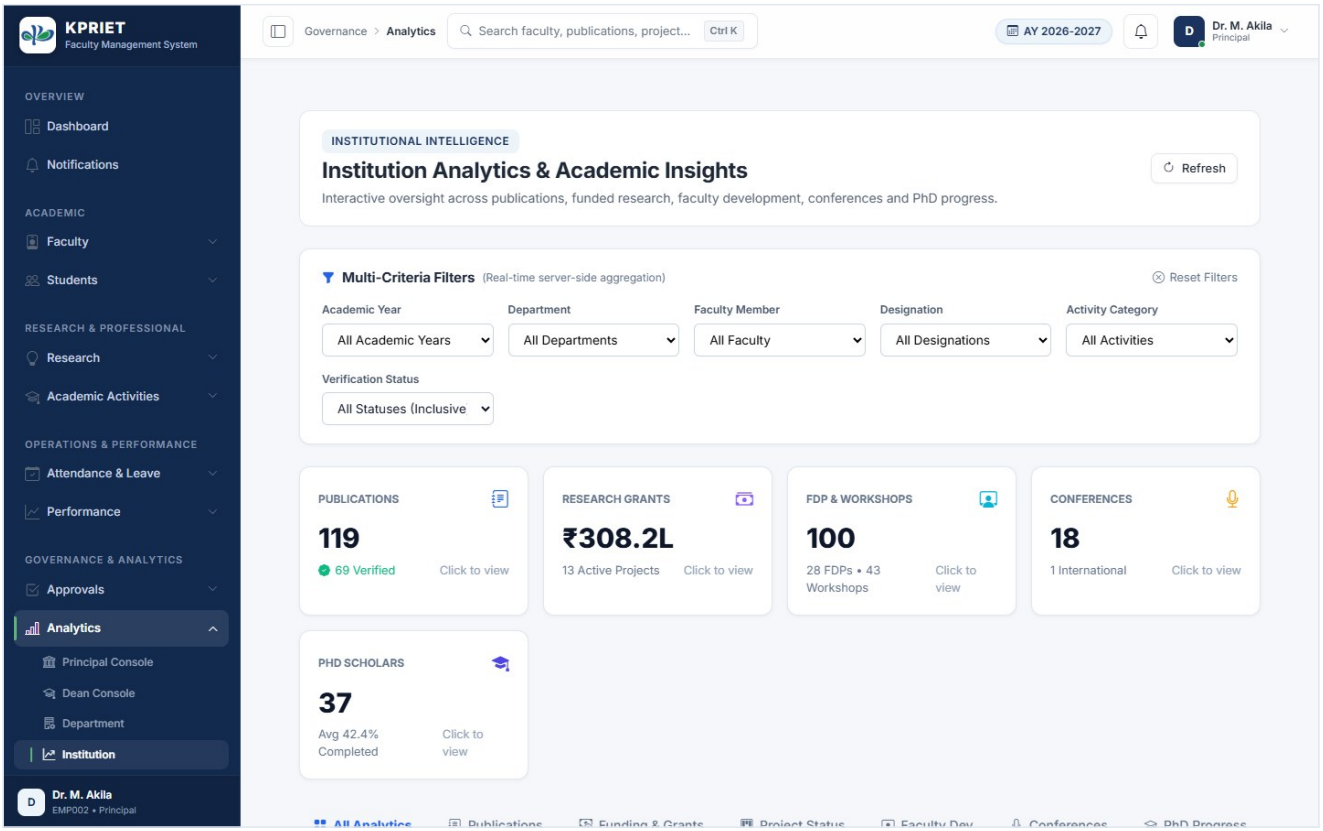
Publications with verification status, filters and DOI/ORCID import



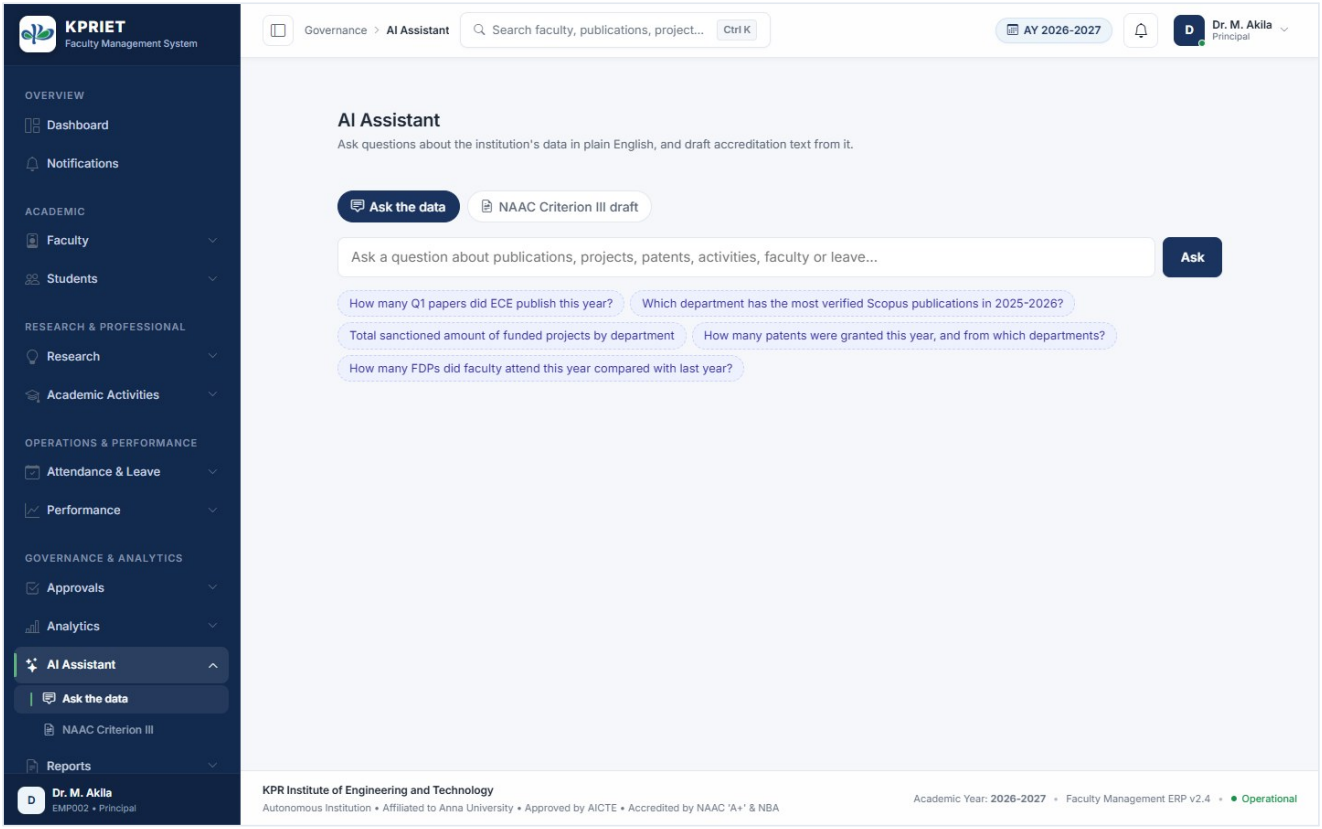
Approval queue for the HOD



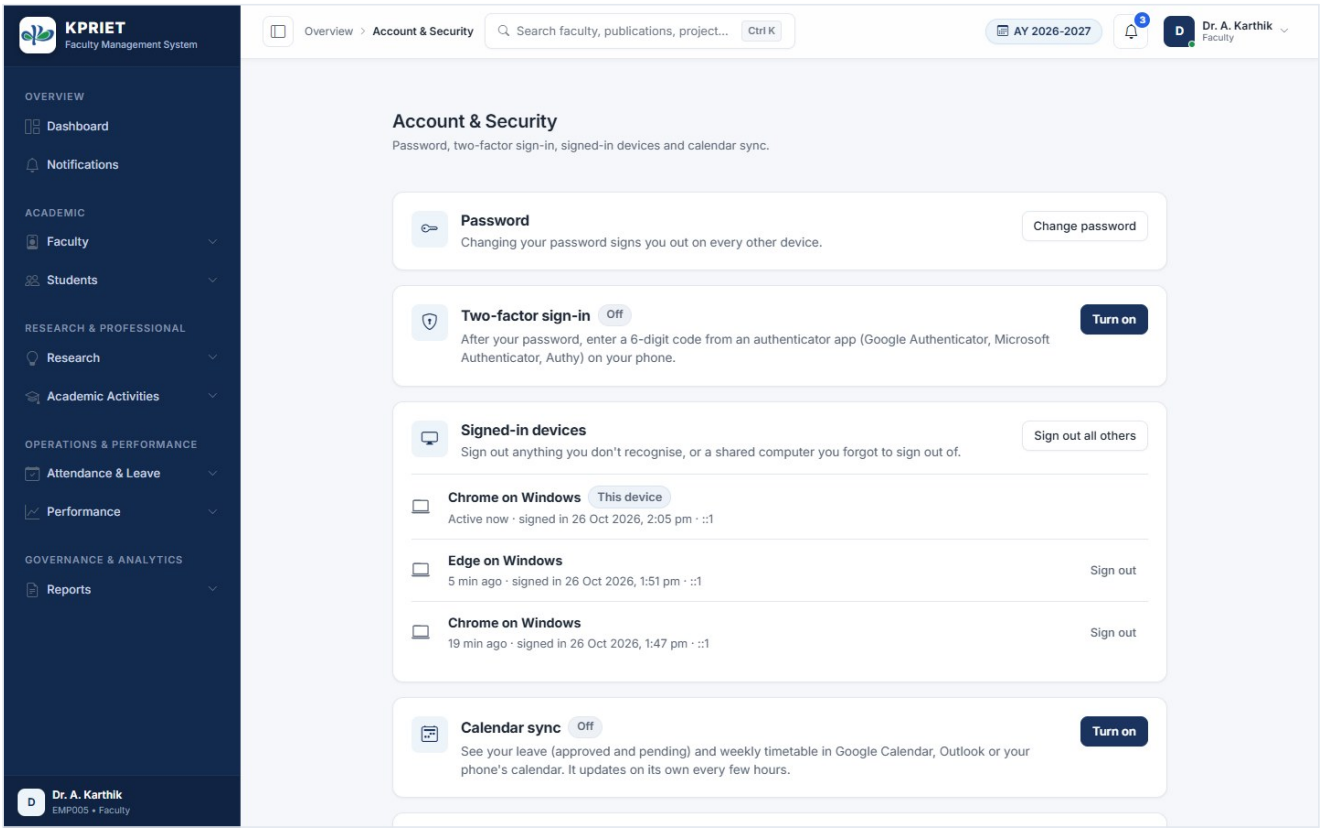
Faculty profile with verification status, research summary and document dossier



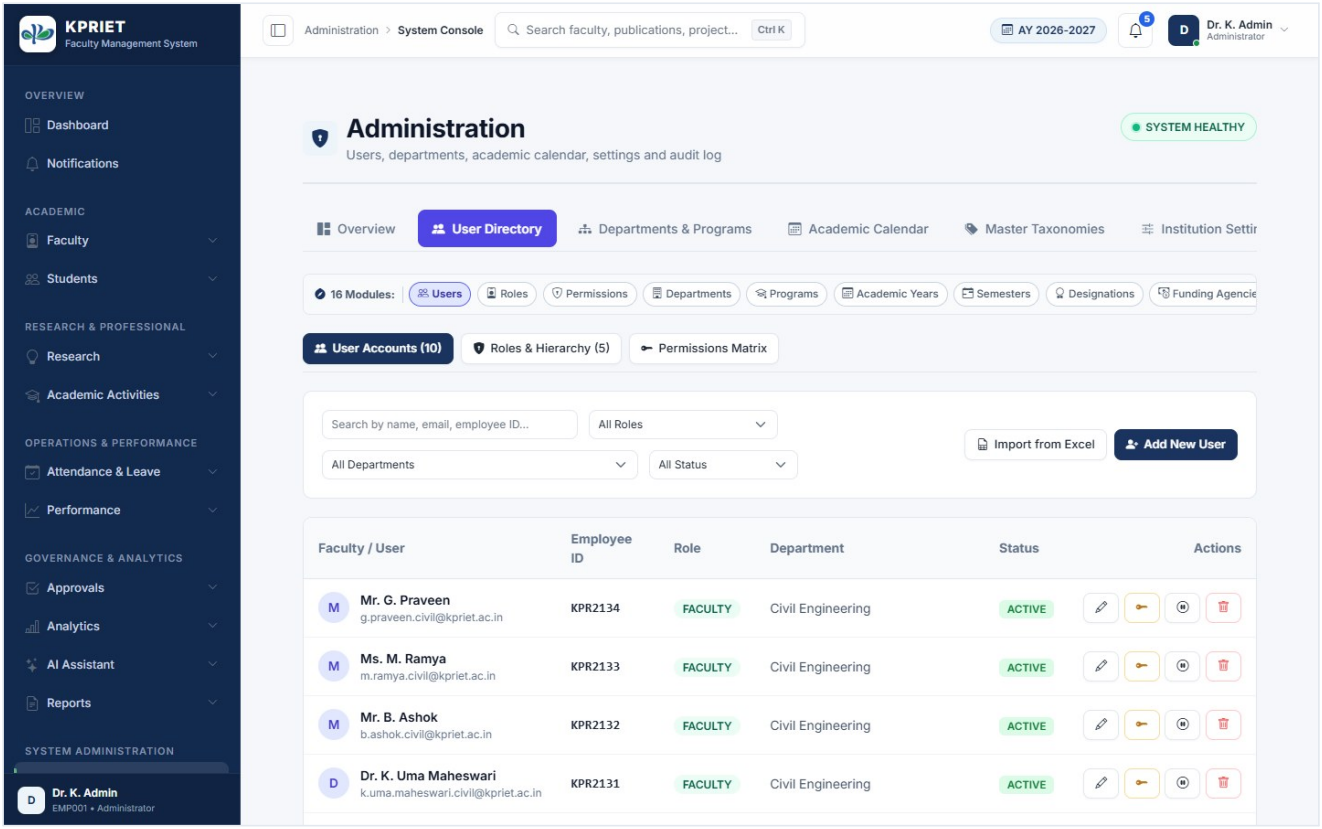
Institution analytics



AI assistant: questions in plain English and NAAC Criterion III drafting



Account & Security: two-step sign-in, signed-in devices, calendar sync



Administration: users, registrations awaiting approval and Excel import

4. Quality and testing

- **36 automated test suites** cover login and sessions, role permissions, data isolation between departments, every module's workflow, file-upload security, account features and AI access rules. All pass on the latest version.
- The tests run on a separate test database, so they never change real data.
- Every screen is checked with TypeScript type checking and a code linter.
- During review we removed several hard-coded figures that were shown as if they were real; every number on the dashboards now comes from the database or shows "—" where data is not recorded.
- Charts that previously drew blank in some browsers were fixed across the dashboards and analytics pages.

Technology used

Layer	Technology
Web application	React 19, TypeScript, Vite, TanStack Query, Recharts
Server (API)	Node.js, Express 5, TypeScript, Zod validation
Database	MongoDB
Security	JWT in secure cookies, CSRF protection, rate limiting, bcrypt, authenticator (TOTP) two-step sign-in
AI	Google Gemini (only summary data is sent for questions and reports)
Deployment	Docker (database, API and web server), with backup and restore scripts

5. Remaining work and next steps

The items below are needed to move from the development environment to daily use at the college.

#	Item	What is needed	From
1	**Production deployment**	A college server or cloud machine with Docker, a web address (for example fms.kpriet.ac.in) and an HTTPS certificate.	KPRIET IT + developer
2	**Email delivery**	Mail server details (SMTP) for a no-reply college mailbox, so password resets, registration codes and approval emails are sent.	KPRIET IT
3	**Real data**	Create the real administrator, departments and programmes; import faculty from Excel; remove the sample data.	Admin office + developer
4	**User acceptance testing**	A trial with a few faculty, HODs, the Dean and the Principal; collect feedback and make changes.	KPRIET + developer
5	**AI key and data policy**	An AI key from the college's own Google account and approval of what data may be sent to the AI service.	KPRIET management
6	**Academic year switcher**	Turn the year badge in the top bar into a selector that sets the year on every page (currently a label).	Developer
7	**Teaching module**	Decide when to switch it on; then review and test it with real timetable data.	KPRIET
8	**Code repository**	Grant repository access so the code can be pushed to the project GitHub repository.	Repository owner
9	**Backups and monitoring**	Schedule daily backups on the server and keep a copy off the server; basic uptime monitoring.	KPRIET IT + developer
10	**Training and user guide**	Short guides and a training session for each role.	Developer

Suggested order

- **Week 1:** server, web address and HTTPS; email set-up; deploy and create the real administrator.
- **Week 2:** load departments and faculty (Excel import); trial with a small group; fix feedback.
- **Week 3:** training for each role; go live; schedule backups.

This report reflects the state of the system on 26 October 2026.